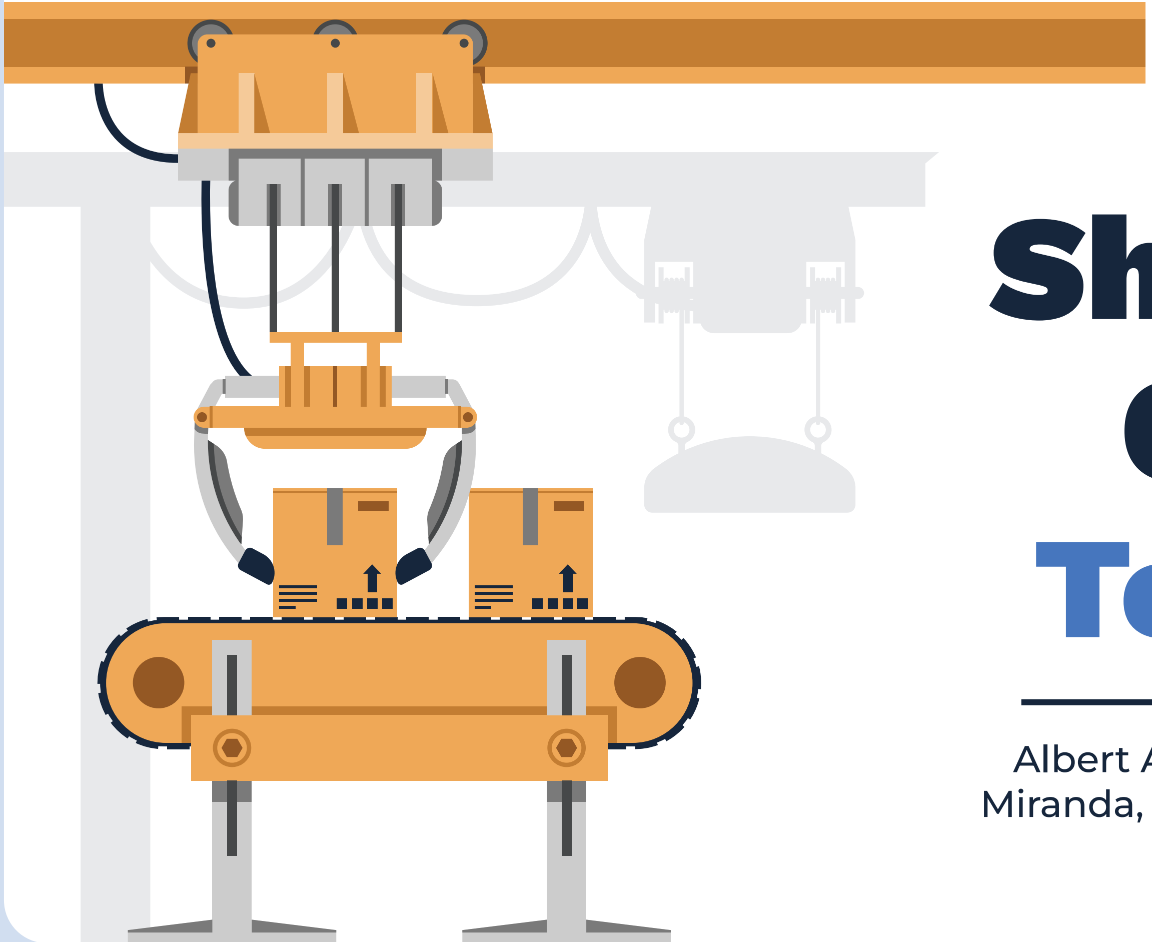


Shaq and Onion

Team 32

Albert Alvarez, Ezekiel Adeleke, Jon Miranda, Nicolas Liu, Thomas Lee-Trejo

Our Goal

- 25 YEARS OF MAE 3!!
- Revamping the original MAE 3 contest
- Putting rings on the Sun God Statue!!

Different Levels equals different point values

- Scoring potential = 1240
- Average points = 930

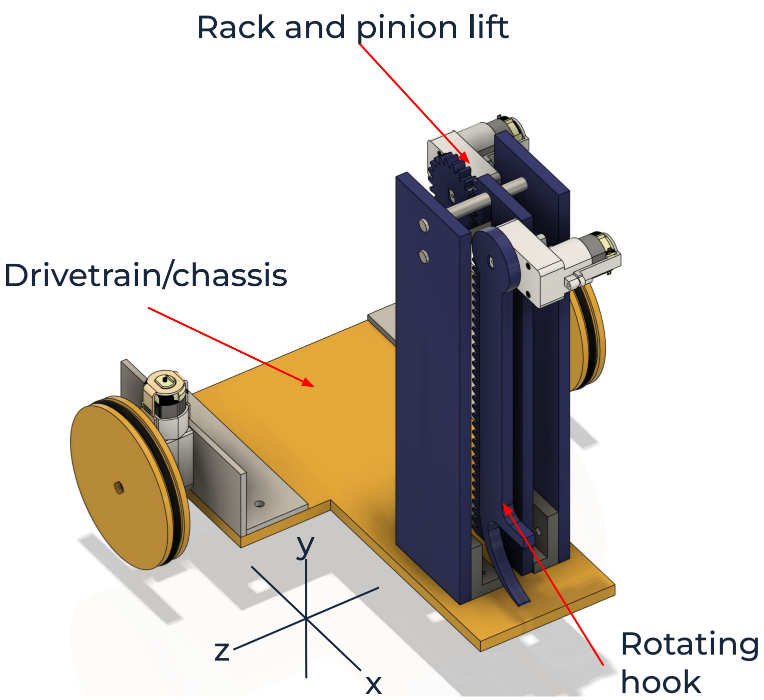
The Robot

As a group, we decided to use two components for the lifting mechanism.

- Rack and Pinion for movement in the z-axis (up and down)
- Rotating about x and y axis to gain more height and change position of hook in the x and y direction (depending on orientation of the robot)

Movement not related to lifting.

- Two motors with wheels and o-rings attached. Motors are controlled separately allowing for ease of turning and more power



How we planned as a team

Team 32 (Shaq and Onion)

Review of Last Week's Progress: How close was team to objectives specfied last week, and what adjustments should be made to planning.

We were able to get things done when needed to as our drive train was our working powered con

Attendance and participation in office hours is definitely not balanced right now, could use some

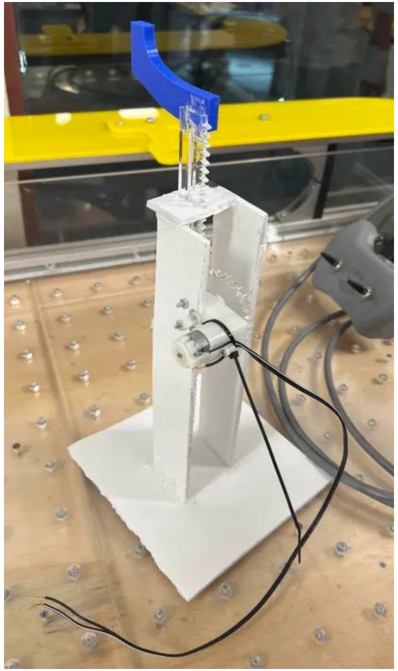
GROUP NAME: Chicken Jockey

DATE: 6/6/25

PHASE ONE: WEEK 5, WEEK 6, WEEK 7 — PHASE TWO: WEEK 8, WEEK 9, WEEK 10 (M T W R F)

	TASK TITLE	TASK OWNER(S)	START DATE	DUE DATE	DURATION (DAYS)	PCT OF TASK COMPLETE
1	Concept Generation	All	4/25/25	5/2/25	7	100%
2	Risk Reduction Testing	All	5/2/25	5/9/25	7	100%
3	Assignment 5C	All	4/25/25	5/2/25	7	100%
4	3D CAD Arms and extra attachments	Nicolas and Jon	5/2/25	5/16/25	14	85%
5	3D Printing and Laser Cutting	Tommy and Albert	5/5/25	5/19/25	14	85%
6	Robot Final Project Presentation Prep	All	5/20/25	6/5/25	15	40%
7	Soldering	Nicolas	5/5/25	5/19/25	14	100%
8	CAD Drive Train	Tommy and Albert	5/11/25	5/17/25	6	100%
9	Building drive train	All	5/12/25	5/17/25	5	100%
10	ReCAD lifting mechanism	Nic and Ezekiel	5/12/25	5/18/25	6	100%
11	Rebuild lifting mechanism	All	5/16/25	5/19/25	3	100%
12	ReCAD ring mechanism	Jon and Ezekiel	5/11/25	5/19/25	8	100%
13	Rebuild ring mechanism	All	5/11/25	5/19/25	8	100%
14	Do analysis for new lifting mechanism	Nicolas	5/21/25	5/22/25	2	100%
15	Create uprights for shafts	Jon	5/20/25	5/22/25	2	100%
16	Attach L-brackets to Drive Train	Jon and Nicolas	5/22/25	5/23/25	1	100%
17	Score 60 points with our robot	All	5/23/25	5/23/25	0	100%
18	FTPE Drive Train	Albert and Thomas	5/25/25	5/31/25	6	30%
19	FTPE Rack and Pinion	Jon and Nicolas	5/25/25	5/31/25	6	80%
20	FTPE Hook	Ezekiel	5/25/25	5/31/25	6	70%
21	Rebulid Columns out of acrylic	Nicolas	5/25/25	5/29/25	4	100%
22	Laser Cut rod spacers	Jon	5/29/25	5/29/25		100%
23	Rebulid Drive Train out of acrylic	Jon	5/26/25	5/29/25	3	100%

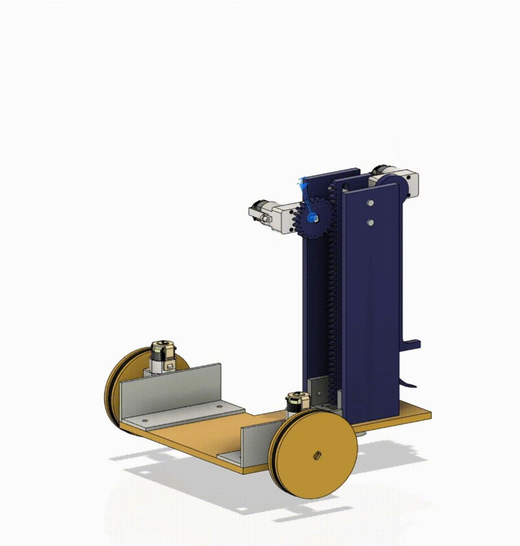
Scheduling: utilizing Gantt charts and communicating availability



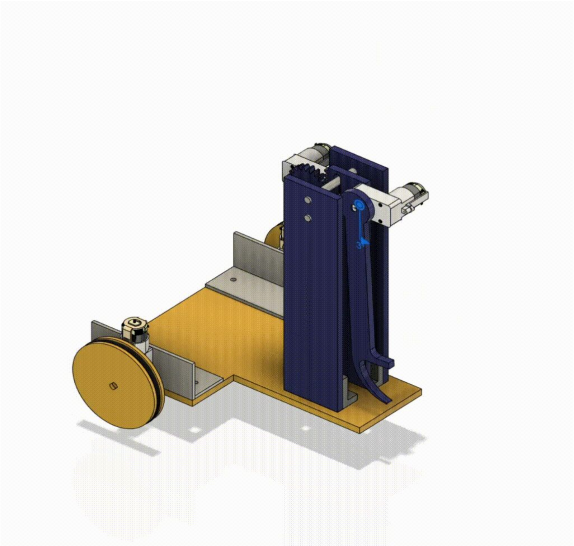
Risk reduction: reliability of lifting mechanism and concept generation

Preview of Motion

Rack and Pinion



Rotating Hook



Drive Train

Breakdown of Wheel

- One 3D printed spacer
- One inner wheel for the medium o-ring
- Two outer wheels to hold o-ring in place

Motors used

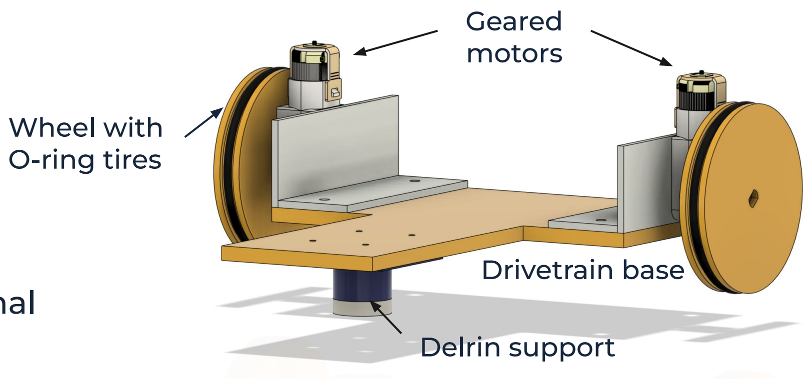
- Geared motors, one per side allowing optimal movement

Pros to having two motors

- Increased mobility and ease of turning (making one wheel go forward and the other go backwards)
- More power than just one motor

Chassis

- T-shaped drivetrain base
- Delrin support near front keeps robot level, choice of Delrin adds minimal friction



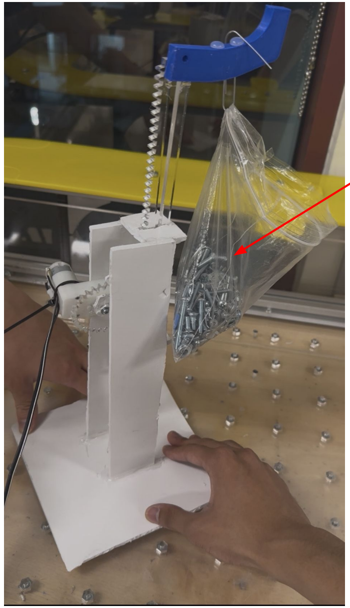
Lifting Mechanism

First component: Rack and Pinion

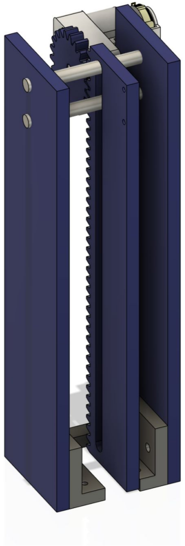
- 90 degree geared motor for ease of placement
- Risk reduction prototype:
  - Made of scrap acrylic
- Final robot design:
  - Made with blue acrylic and held together by metal rods, acrylic columns, and 3D printed U-bracket

Pros

- Manufacturing is relatively easy given the CAD library provided by MAE 3
- Can fine tune the specific height without having to move the actual robot



Risk reduction prototype

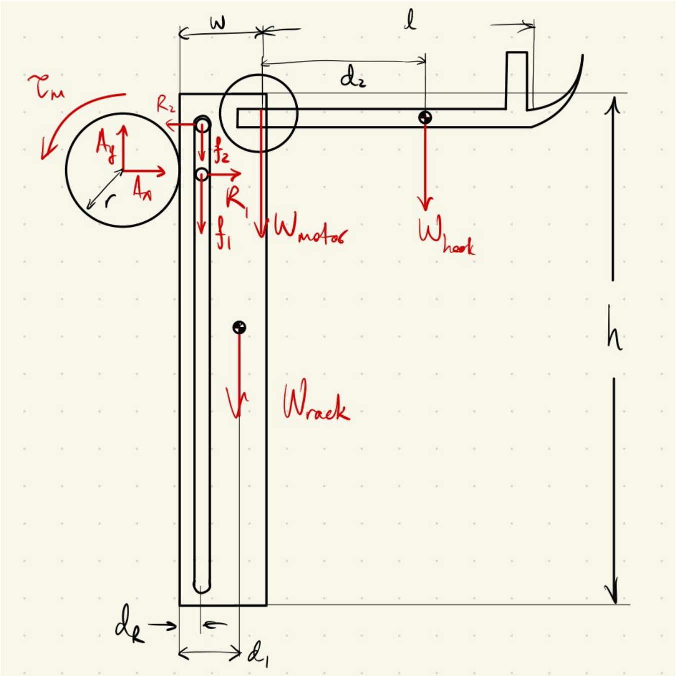


Cadded model of final product

FTPE Analysis of Rack and Pinion

Force-Torque Analysis

External FBD:



Assumptions:

- Friction only occurs at aluminum shafts
- Gear teeth make 30-deg angle from vertical

Theoretical results:

- Assembly can lift equivalent of 1.47 kg above rack
- Moment arm of hook will decrease this where load is applied

Experimental results:

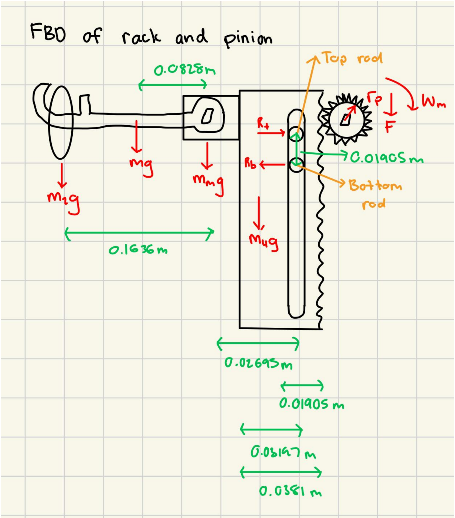
- Lifted much less, around 0.15 kg
- Likely due to high friction from added bearings, which are not accounted for in this calculation

For 1 ring:
Theoretical FoS: 183.75
Experimental FoS: 18.75

FTPE Analysis of Rack and Pinion

Power and Energy Analysis: How fast can the rack go?

External FBD:



Measured value: 3.6 inch/sec
Calculated value: 4.4 inch/sec
Percent error: 22.4%

Assumptions:

- Motor is operating at max power
- Neglecting friction between teeth
- Rack goes perfectly straight up
- Quasi-static analysis

Calculate the force generated by the pinion when motor is operating a max power

- Force generated by pinion = $\frac{\frac{1}{2} * \tau_{stall}}{r_{pinion}}$

Calculate the reaction force to get the force of friction

- Reaction force
  - $R_b = \frac{dist_{motor}(W_{motor}) + (dist_{ring} + dist_{motor})(W_{ring}) + (dist_{hook} + dist_{motor})(W_{hook})}{dist_{rod}}$

Calculate the total work done on the system

- Total work =
  $F(h) + 2(R_b)(\mu_{alum})(h) + (m_{motor} + m_{rack} + m_{ring} + m_{hook})(g)(h)$

Calculate the time it takes for the motor to generate that much energy

- Solve for time = $\frac{Work}{P_{max}}$

Divide change in height by time to get velocity

- $v_{rack} = \frac{h}{t}$

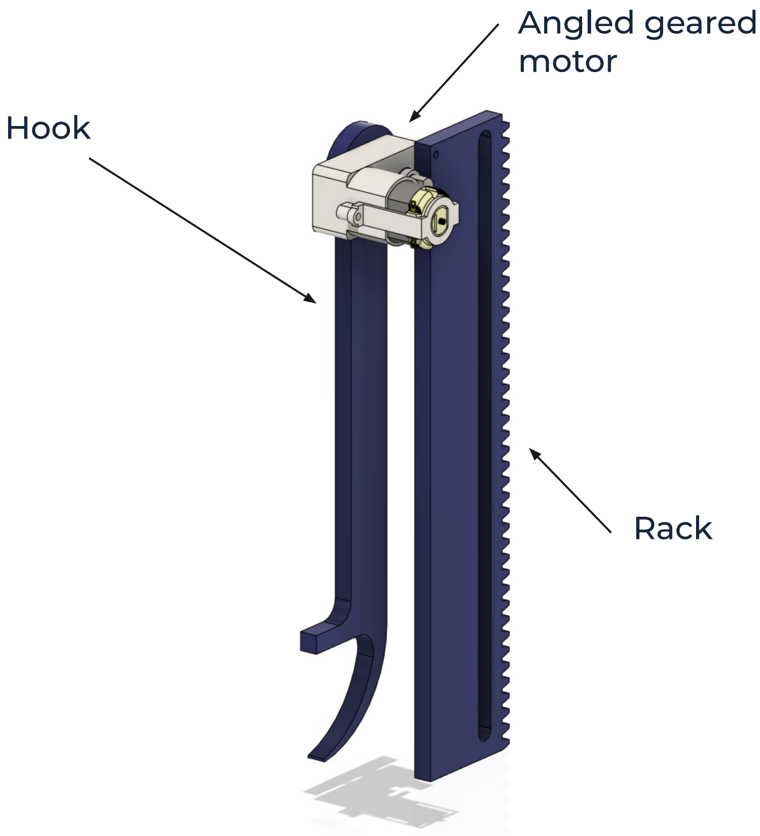
Lifting Mechanism

Second Component: Rotating Hook

- 180 degrees of movement allowed due to the restriction caused by the placement of the metal rods to keep the rack aligned
- 8.5 inches in length to allow for extra reach
- Directly screwed onto the top of the rack with two 4-40 screws
- Also acts as primary method of retrieving rings and scoring (more on exact method on scoring section)

Pros to using a rotating hook

- Easy to manufacture and doesn’t require too much precision
- While simple, it’s very effective

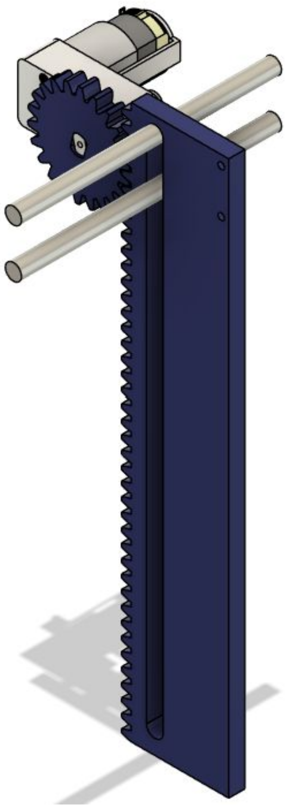


Ezekiel

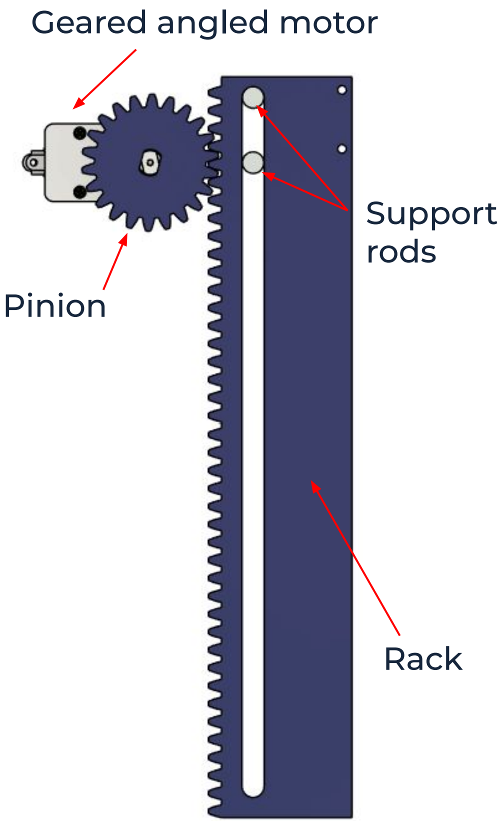
Rack and Pinion and Hook Works!

- Scoring details:
    - Max score: 1240
    - Average score: 930
    - Most common score: 930

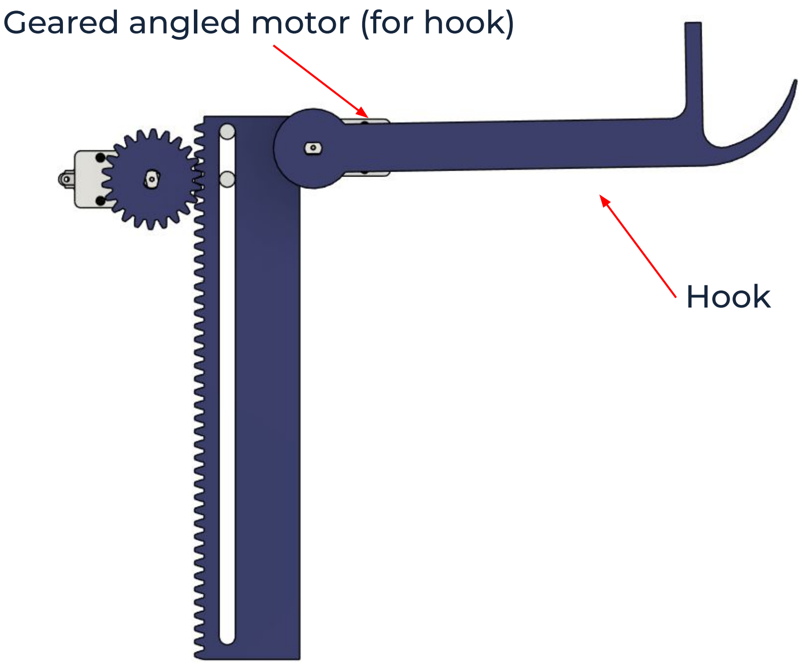
- Video of our robot scoring →



Isometric View

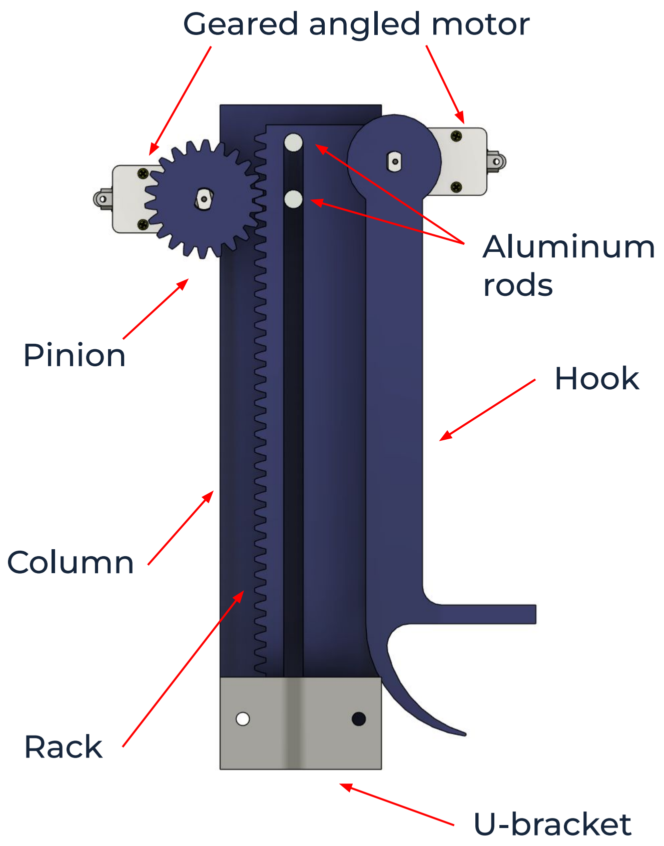


Back View (rack and pinion only)

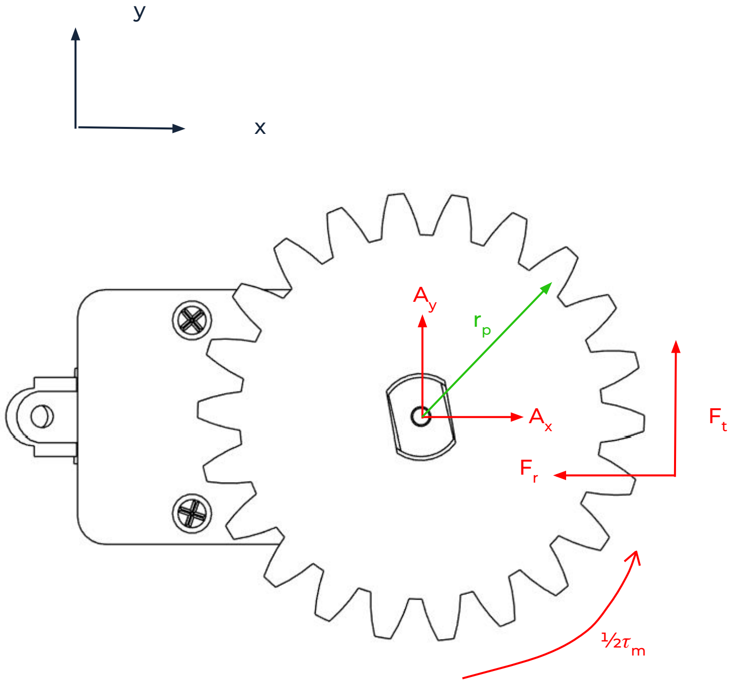


Back View for analysis

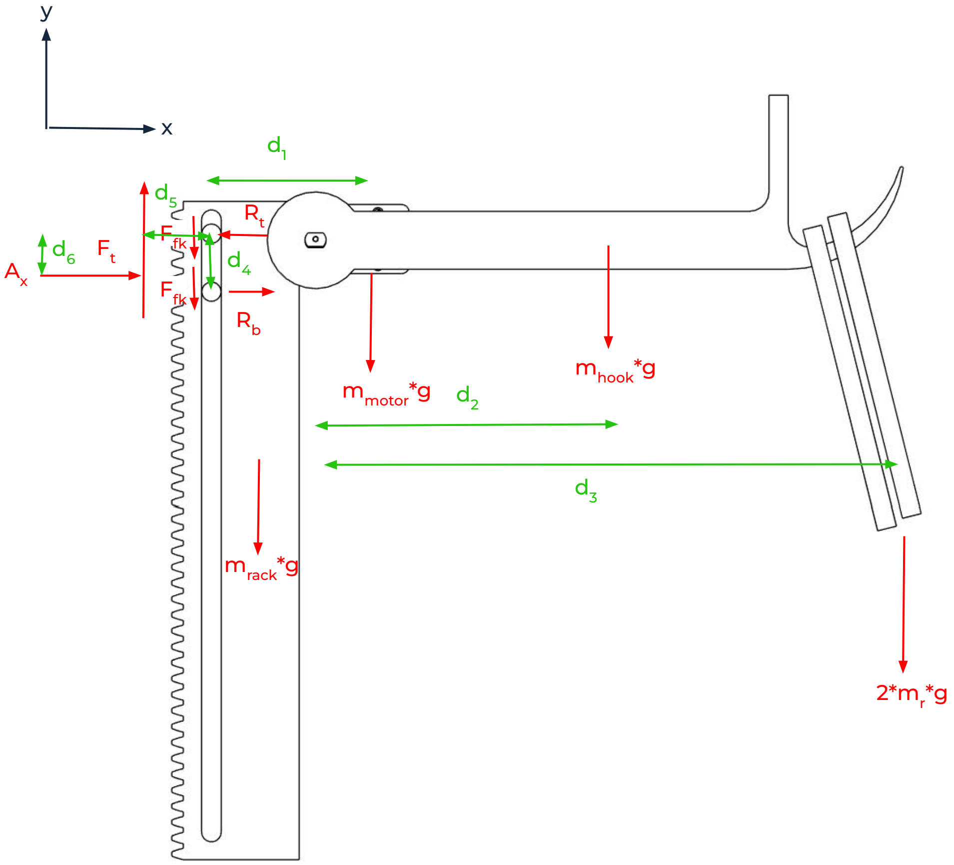
Isometric View



Side View



y
x
A_y
r_p
A_x
F_t
F_r
½τ_m



y
x
d_1
d_5
R_t
F_fk
d_6
F_t
A_x
d_4
F_fk
R_b
m_hook*g
m_motor*g
d_2
d_3
m_rack*g
2*m_r*g